

BRANDON, MB, Canada

WMO: 711400

Lat: 49.9033N

Lon: 99.9456W

Elev: 409

StdP: 96.50

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.7	(c) -30.2	(d) -37.3	(e) 0.1	(f) -32.5	(g) -34.8	(h) 0.1	(i) -30.1	(j) 13.1	(k) -11.3	(l) 11.3	(m) -11.5	(n) 3.9	(o) 280	(p) 0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	30.0	20.7	28.2	19.5	26.6	18.6	22.2	27.8	21.0	26.5	19.8	24.9	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.9	25.7	19.0	14.5	23.9	17.8	13.4	22.7	67.6	27.9	62.7	26.7	58.5	24.9	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.7	(c) 8.5	(e) -36.1	(f) 34.0	(g) 2.6	(h) 1.8	(i) -37.9	(j) 35.3	(k) -39.5	(l) 36.3	(m) -40.9	(n) 37.3	(o) -42.8	(p) 38.6		
(a) 11.3	(b) 9.7	(c) 8.5	(e) -36.2	(f) 24.8	(g) 2.6	(h) 1.7	(i) -38.0	(j) 26.0	(k) -39.5	(l) 27.1	(m) -41.0	(n) 28.0	(o) -42.9	(p) 29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-16.3	-14.3	-6.6	3.4	10.4	16.2	18.8	17.8	12.7	4.6	-5.0	-13.2
	DBStd	13.85	8.24	7.68	7.73	6.00	4.72	3.59	3.02	3.42	4.41	4.94	6.63	7.66
	HDD10.0	3643	815	679	516	208	52	2	0	0	22	179	451	719
	HDD18.3	5899	1073	912	773	448	248	82	31	51	176	427	701	977
	CDD10.0	893	0	0	1	9	66	188	273	243	102	11	0	0
	CDD18.3	108	0	0	0	0	3	18	46	36	6	0	0	0
	CDH23.3	1398	0	0	0	3	74	224	488	481	122	7	0	0
	CDH26.7	353	0	0	0	0	16	49	114	147	26	1	0	0
Wind	WSAvg	4.2	4.2	4.2	4.3	4.5	4.7	4.1	3.5	3.6	4.0	4.4	4.2	4.1
Precipitation	PrecAvg	471	21	18	22	37	52	68	73	70	53	23	16	20
	PrecMax	722	47	63	55	113	124	192	160	222	112	74	41	45
	PrecMin	320	3	0	3	0	6	10	12	3	7	1	2	6
	PrecStd	96	11	16	14	27	31	37	42	55	29	22	11	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	3.2	14.2	23.0	28.9	30.8	31.6	33.4	29.6	23.4	13.2	4.4
		MCWB	1.1	1.4	8.1	12.7	16.5	20.8	22.6	21.3	18.8	14.3	7.5	1.8
	2%	DB	0.5	1.1	8.9	19.2	25.4	27.9	29.4	30.2	26.6	18.8	9.0	1.5
		MCWB	-0.6	0.0	5.4	10.5	14.3	18.5	21.5	20.4	17.4	11.7	4.8	-0.1
	5%	DB	-1.8	-0.5	5.1	16.6	22.5	25.8	27.8	28.0	24.3	15.6	5.8	-1.0
		MCWB	-2.6	-1.3	2.9	9.2	13.1	17.4	20.6	19.4	16.3	9.9	3.3	-1.9
	10%	DB	-4.2	-2.8	2.7	13.5	19.8	24.0	26.1	26.0	21.4	12.7	3.3	-2.8
		MCWB	-4.8	-3.6	1.2	7.7	11.9	16.8	19.5	18.4	14.6	8.6	1.6	-3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.7	8.9	13.4	18.2	23.3	24.5	23.6	20.7	14.8	8.3	2.1
		MCDB	2.6	2.9	13.6	21.0	25.3	29.1	29.1	29.6	26.5	22.5	12.3	4.0
	2%	WB	-0.5	0.3	5.5	11.3	15.7	20.4	22.6	21.7	18.6	12.5	5.5	0.0
		MCDB	0.5	1.1	9.0	18.0	21.8	25.6	27.9	28.1	24.4	17.4	8.4	1.4
	5%	WB	-2.5	-1.3	3.0	9.6	14.3	18.8	21.4	20.4	17.2	10.7	3.3	-1.8
		MCDB	-1.7	-0.5	5.0	15.7	20.4	23.7	26.6	26.4	22.4	14.6	5.5	-1.1
	10%	WB	-4.8	-3.7	1.3	8.0	12.7	17.6	20.2	19.2	15.5	9.0	1.7	-3.5
		MCDB	-4.2	-2.8	2.5	13.1	18.5	22.3	25.1	24.3	20.2	12.6	3.3	-2.8
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.1	9.9	11.8	13.0	11.6	12.3	13.6	13.2	11.1	9.2	9.0
		MCDBR	11.0	9.9	11.4	17.3	17.9	15.0	14.6	17.1	18.0	16.9	13.3	10.7
		MCWBR	10.2	8.9	8.4	10.0	9.1	7.7	7.9	8.6	9.8	10.3	9.6	9.6
	5% WB	MCDBR	10.8	9.5	11.1	16.0	14.6	12.6	13.4	14.8	15.5	15.3	12.4	10.3
		MCWBR	10.0	8.6	8.3	9.7	8.2	7.2	7.9	8.1	9.0	9.7	9.2	9.2
Clear-Sky Solar Irradiance	taub	0.260	0.265	0.305	0.344	0.357	0.367	0.381	0.383	0.347	0.290	0.262	0.235	
	taud	2.271	2.283	2.308	2.348	2.373	2.384	2.352	2.340	2.418	2.547	2.472	2.378	
	Ebn at Noon	790	886	904	901	900	890	870	851	849	857	801	787	
	Edh at Noon	69	88	104	113	115	115	117	113	94	69	57	53	
All-Sky Solar Radiation	RadAvg	1.43	2.60	3.92	5.04	5.65	5.96	6.34	5.19	3.80	2.29	1.27	1.02	
	RadStd	0.11	0.20	0.30	0.35	0.45	0.41	0.29	0.39	0.30	0.29	0.17	0.11	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (21 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) -0.66	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page